

Instruções peppo sadasdsdasdsdasdsadsadsads

1. Acesse os links dos simuladores e realize a atividade.
 2. Em local determinado, deve-se colar a foto do printScreen da tela de resultados e foto dos cálculos no caderno.
 3. Salve o arquivo em formato pdf nomeando-o com seu nome.
 4. NÃO SERÁ ACEITO: arquivo em WORD e sem nome do arquivo.
 5. Arquivo realizado fora do prazo (update) será considerado como não entregue.
 6. Encaminhe o arquivo para o link abaixo
- https://drive.google.com/drive/folders/12ug5_6uOyeXrimcyeEErTn-Q7kyHmtXm?usp=sharing

Simulador1

link: <https://thephysicsaviary.com/Physics/APPrograms/AvgKEandTProblem/>

Foto dos cálculos Simulador 1

Reado Antonio.

$K_B = 1,38 \cdot 10^{-23} \text{ J/K}$
 $T = 112 \text{ K}$

$KE = \frac{3}{2} \times (1,38 \cdot 10^{-23} \text{ J/K}) (112 \text{ K})$

$KE = 2,3181 \cdot 10^{-21} \text{ J}$

$m = 39,94 \cdot 1,66054 \cdot 10^{-27} \text{ kg}$
 $m = 6,6321 \cdot 10^{-26} \text{ kg}$

Resposta:
$KE = 2,318 \cdot 10^{-21} \text{ J}$
$v = 264,35$

$\sqrt{\frac{3 \cdot (1,38 \cdot 10^{-23} \text{ J/K}) \cdot (112 \text{ K})}{6,6321 \cdot 10^{-26} \text{ kg}}}$

$\sqrt{\frac{4,63488 \cdot 10^{-21}}{6,6321 \cdot 10^{-26}}}$

$\sqrt{69880,6}$

$v_{rms} = 264,35 \text{ m/s}$

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printScreen Simulador 1

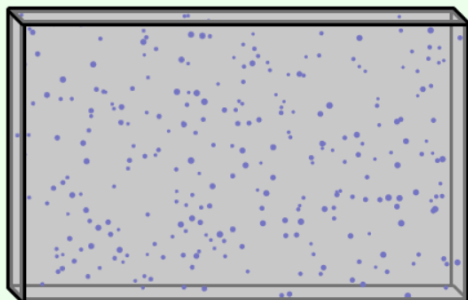
Average KE and rms Problem

Pedro Antonio (Average KE and rms Problem)

Calculations Correct

Argon (39.94 amu)

112 K



Average KE (J):

Predicted = 2.318×10^{-21} , Actual = 2.32×10^{-21}

v (m/s)

Predicted = 264.35, Actual = 264.5

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Pesquisar



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16/03/2026

Simulador2

link: <https://thephysicsaviary.com/Physics/APPrograms/TemperatureBasedOnSpeed/>

Foto dos cálculos Simulador 2

Redo Antonio.

$$1 \text{ amu} = 1.66054 \cdot 10^{-27} \text{ kg}$$

$$m = 20.179 \text{ amu} \cdot 1.66054 \cdot 10^{-27} \text{ kg/amu}$$

$$m = 3.3517 \cdot 10^{-26} \text{ kg}$$

$$KE = \frac{1}{2} m v^2$$

$$\frac{1}{2} \cdot (3.3517 \cdot 10^{-26} \text{ kg}) \cdot (500 \text{ m/s})^2$$

$$\frac{1}{2} \cdot (3.3517 \cdot 10^{-26}) \cdot 250000$$

$$4.1896 \cdot 10^{-21} \text{ J}$$

$$KE_{\text{avg}} = \frac{3}{2} kT$$

$$(1.38 \cdot 10^{-23} \text{ J/K})$$

$$T = \frac{2 \times KE_{\text{avg}}}{3k}$$

$$T = \frac{2 \times (4.1896 \cdot 10^{-21} \text{ J})}{3 \times (1.38 \cdot 10^{-23} \text{ J/K})} = \frac{8.3792 \cdot 10^{-21}}{4.14 \cdot 10^{-23}} = T = 202.4$$

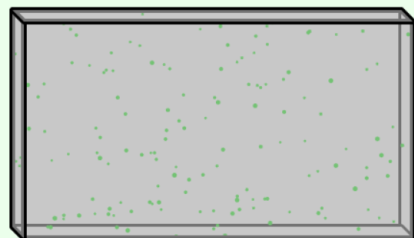
$$KE_{\text{avg}} = 4.1896 \cdot 10^{-21}$$

$$T = 202.4$$

512

printScreen Simulador 2

Temperature Based on Speed



Neon (20.179 amu)



Pedro Antonio

Temperature Based on Speed

Calculations Correct

Givens

$$v(\text{avg}) = 500 \text{ m/s}$$

$$m = 3.35\text{e-}26 \text{ kg}$$

Calculations

$$\text{KE}(\text{avg}) \text{ (J): Predicted} = 4.1896\text{e-}21, \text{ Actual} = 4.181\text{e-}21$$

$$T \text{ (K): Predicted} = 202.4, \text{ Actual} = 202$$

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isolado

Pesquisar

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